

Venkat Mandalapu

Software Engineer

TX, USA | ☎ +1 469-820-0203 | ✉ Venkatk1315@gmail.com | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Software Engineer with over **5 years of experience** developing enterprise applications in **healthcare, fintech, and global marketplace domains**, delivering systems that process more than **45M transactions annually** and manage over **9TB of data**. Skilled in full-stack development with **Java, Spring Boot, React.js, Angular, Next.js, and .NET Core**, and experienced in designing APIs using **REST, SOAP, GraphQL, and API Gateway**. Hands-on expertise in **payment gateway integrations, Postman-based API validation, microservices, CI/CD pipelines, Docker, Kubernetes, and cloud platforms** including **AWS and Azure**. Strong background in **system design, debugging, performance optimization**, and consistent delivery of **secure, scalable, and compliance-ready solutions**.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, TypeScript, Python, C++, C#, Go (Golang), SQL, HTML5, CSS3

Frameworks & Libraries: Spring Boot, Spring MVC, Hibernate, React.js, Angular, Next.js, AngularJS, Vue.js, Node.js, Express.js, GraphQL, Redux, React Query, Tailwind CSS, Material UI, Bootstrap, Flutter, NumPy, Pandas, Matplotlib

Web Technologies: REST APIs, SOAP, GraphQL, WebSockets, JSON, XML, jQuery, AJAX, **API Gateway**, Secure Payment Integrations

Databases: PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, DynamoDB, SQLite, Cassandra, BigQuery, Snowflake, Firebase, Hadoop, Kafka, Flink

Cloud & DevOps: AWS (EC2, Lambda, S3, RDS, API Gateway, CodePipeline, CodeDeploy, Route 53, CloudWatch), Azure (Blob Storage, Data Lake, Cosmos DB, Functions), Docker, Kubernetes, Jenkins, GitHub Actions, GitLab CI/CD, Bitbucket, Terraform, CloudFormation, Maven, Gradle

Testing & QA: JUnit, Mockito, Selenium, Cypress, PyTest, **Postman**, Swagger, Chrome DevTools, Node.js Profiler

Engineering Skills: System Design, Microservices Architecture, **Scalable Application Development, Authentication & Authorization, CI/CD Pipelines**, Cloud-Native Deployments, **Debugging & Troubleshooting, Performance Optimization, Agile Development**

Tools: Figma, Adobe XD, Microsoft Visio, IntelliJ IDEA, Eclipse, Visual Studio Code, Linux/Unix, Windows, macOS, Red Hat

PROFESSIONAL EXPERIENCE

Eli Lilly and Company — Java Developer

📍 Dallas, TX

Apr 2025 – Present

- Designed and developed enterprise-grade modules in **Java, Spring Boot, and Hibernate** that securely processed more than **48M clinical trial records** across distributed systems.
- Built and integrated **REST, SOAP, and GraphQL APIs** with **Oracle** and **AWS RDS** through **API Gateway** to enable global research data access and secure interoperability.
- Implemented **authentication and authorization layers** using **OAuth2 and JWT** to ensure **HIPAA and FDA compliance**.
- Developed **secure payment gateway integrations** for licensed billing workflows in clinical systems.
- Managed deployments with **Jenkins, Docker, Kubernetes, and AWS CodePipeline**, completing more than **11K automated releases**.
- Enhanced observability and reliability with **CloudWatch, Splunk, and Postman API collections**, reducing downtime by **25%**.

Molina Healthcare — Software Developer

📍 USA

Jul 2024 – Apr 2025

- Built **HIPAA-compliant healthcare applications** with **Spring Boot, React.js, Redux, React Query, and Tailwind CSS** that served more than **850K active members** and processed **12M annual claims**.
- Developed **FHIR and HL7 APIs** with **Azure Blob Storage, Functions, and Cosmos DB** integrated through **API Gateway** for secure clinician data exchange.
- Engineered lightweight microservices in **Go (Golang)** for secure clinician data exchange workflows, improving **API response times by 15%**.
- Optimized **eligibility and claims workflows** to process **7.4TB of daily data**, improving throughput by **30%**.
- Automated **9.8K deployments** across **Jenkins, GitHub Actions, GitLab CI, and Bitbucket** to streamline releases.
- Strengthened service quality with automated testing using **JUnit, Mockito, Cypress, PyTest, Selenium, and Postman**.

Airbnb — Software Engineer

📍 Bengaluru, India

Jun 2019 – Jun 2022

- Developed booking and reservation workflows with **Python Flask, FastAPI, React.js, and Next.js**, scaling to **27M annual transactions** and **1.8M daily requests**.
- Designed **fraud detection pipelines** with **Kafka and Flink** that processed more than **3.2B logs annually** for secure payment systems.
- Built analytics workflows with **NumPy, Pandas, Matplotlib, BigQuery, and Snowflake**, managing **11TB of transactional datasets**.
- Migrated backend services to **GCP** with **Terraform and Kubernetes**, scaling deployments to support global traffic surges.
- Redesigned **payment APIs** with **GraphQL, Node.js, and Express.js** integrated with **payment gateways** supporting **640K daily sessions**.
- Built scalable backend services in **Go (Golang)** integrated with **Kafka and Flink** to process high-volume transactional data across global payment systems.
- Enhanced monitoring and reliability with **Datadog, Splunk, and Postman API test suites**.

Dell Technologies — Frontend Developer	
📍 Hyderabad, India	May 2018 – Jun 2019
- Developed responsive enterprise UIs with Angular, AngularJS, React.js, and Next.js, improving accessibility for more than 220K enterprise users worldwide. - Integrated REST and SOAP APIs through API Gateway to securely process 12M records across distributed systems. - Built modular UI components with Bootstrap, Tailwind CSS, Material UI, and Sass that accelerated the delivery of 48+ enterprise modules. - Conducted cross-browser and API testing with Selenium, Postman, and Chrome DevTools to ensure performance stability across global users. - Collaborated within Agile squads using JIRA, Confluence, and Visio to manage sprints and technical documentation.	

EDUCATION	
Texas A&M University – Commerce, TX	
Master of Science in Computer Science GPA: 3.6 / 4.0	Aug 2022 – May 2024

PROJECTS	
Student Assistant System Jan 2024 – May 2024	
- Built a cloud mentorship platform on AWS with API Gateway, RDS, Lambda, and Amplify using Angular and Spring Boot to support 2.3K authenticated users. - Integrated scheduling workflows and secure payment APIs validated with Postman, enabling 420+ monthly appointments.	
Local Train Ticketing System Aug 2023 – Dec 2023	
- Developed an SQL-based ticketing system with SOAP and REST APIs routed through API Gateway and tested with Postman, processing 38K monthly bookings. - Integrated secure payment workflows and automated reporting for real-time seat allocation and transaction auditing.	
Genome Data Processing with MapReduce Jan 2023 – May 2023	
- Designed Hadoop MapReduce pipelines for genomic and e-commerce datasets totaling 7.8TB, processing more than 1.6B records. - Improved clustering throughput by applying Self-Organizing Maps (SOM) and fuzzy logic models supported by NumPy and Pandas.	